

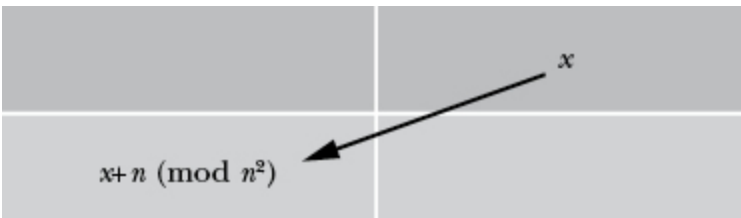
```

square[center_i][center_j + 1] = n**2 + 1 - n
square[center_i][center_j - 1] = n

```

Specifying the Three Rules

The purpose of Kurushima's algorithm is to fill in the rest of the `nan` entries according to simple rules. We can specify three simple rules that enable us to fill out every other entry, no matter how big the magic square is. The first rule is expressed in [Figure 2-1](#).



[Figure 2-1](#): Rule 1 of Kurushima's algorithm

So for any x in the magic square, we can determine the entry that is situated in this diagonal relationship to x by simply adding n and taking the result $\text{mod } n^2$ (mod refers to the modulo operation). Of course, we can also go in the opposite direction by reversing the operation: subtracting n and taking the result $\text{mod } n^2$.

The second rule is even simpler, and is expressed in [Figure 2-2](#).



[Figure 2-2](#): Rule 2 of Kurushima's algorithm

For any x in the magic square, the entry below and to the right of x is 1 greater than x , $\text{mod } n^2$. This is a simple rule, but it has one important exception: this rule is not followed when we cross from the upper-left half of the magic square to the lower-right half of the square. Another way to say this is that we do not follow the second rule if we are crossing the